WORKSHEET 4.2 : INTERSECTION OF A LINE AND A PLANE

1a) $x = 4 - t$
$y = 6 + 2t$
$z = -2 + t$ $\Big\}$ ① $t \in \mathbb{R}$

② $2x - y + 6z + 10 = 0$

$2(4-t) - (6+2t) + 6(-2+t) + 10 = 0$

$8 - 2t - 6 - 2t - 12 + 6t + 10 = 0$

$2t = 0$

$t = 0$

∴ P.O.I (4, 6, −2)

b) $x = 3 + 4t$
$y = -2 - 6t$
$z = \frac{1}{2} - 3t$ $\Big\}$ $t \in \mathbb{R}$

$3x + 4y - 7z + 7 = 0$

$3(3+4t) + 4(-2-6t) - 7(\frac{1}{2} - 3t) + 7 = 0$

$9 + 12t - 8 - 24t - \frac{7}{2} + 21t + 7 = 0$

$9t + 8 - \frac{7}{2} = 0$.

$18t + 16 - 7 = 0$.

$18t = -9$

$t = -\frac{1}{2}$

∴ $x = 3 + 4(-\frac{1}{2}) = 1$

$y = -2 - 6(-\frac{1}{2}) = 1$

$z = \frac{1}{2} - 3(-\frac{1}{2}) = 2$

∴ P.O.I (1, 1, 2)

c) $x = 5 + t$
$y = 4 + 2t$
$z = 7 + 2t$ $\Big\}$ $t \in \mathbb{R}$

$2x + 3y - 4z + 7 = 0$.

$2(5+t) + 3(4+2t) - 4(7+2t) + 7 = 0$

$10 + 2t + 12 + 6t - 28 - 8t + 7 = 0$

$0t = -1$

no solution

∴ Line is // to plane

Check $\vec{m} \cdot \vec{n}$

$= (1, 2, 2) \cdot (2, 3, -4)$

$= 2 + 6 - 8$

$= 0$ ✓

d) $\left.\begin{aligned} x &= 2-t \\ y &= 14-t \\ z &= 1+t \end{aligned}\right\} t \in \mathbb{R}$

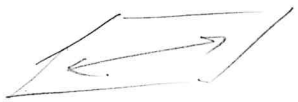
$3x - y + 2z + 6 = 0$

$3(2-t) - (14-t) + 2(1+t) + 6 = 0$

$6 - 3t - 14 + t + 2 + 2t + 6 = 0$

$\underbrace{0t = 0}_{\text{infinite solution}}$

$\therefore$ line on the plane



$\therefore (x, y, z) = (2-t, 14-t, 1+t) \quad t \in \mathbb{R}$

$OR = (2, 14, 1) + t(-1, -1, 1), \; t \in \mathbb{R}$

e) $\left.\begin{aligned} x &= 5 \\ y &= 7+t \\ z &= 3-t \end{aligned}\right\} t \in \mathbb{R}$

$z + 5 = 0$

$(3-t) + 5 = 0$

$t = 8$

$\therefore$ P.O.I $\quad x = 5$

$y = 7 + 8 = 15$

$z = 3 - 8 = -5$

$\therefore$ P.O.I $(5, 15, -5)$

2a) option #1:

$\vec{m} \cdot \vec{n}$ should equal to zero if parallel

$(3, 2, -1) \cdot (3, -4, 1)$

$= 9 - 8 - 1$

$= 0$ ✓

check point from line if coincident (share a point)

$LS = 3(-2) - 4(6) + (5) + 25 \qquad RS = 0$

$= -6 - 24 + 5 + 25$

$= 0$

$LS = RS$ $\therefore$ Line lies on plane

b) option #2

$\left.\begin{array}{l} x = 4+3t \\ y = -1+2t \\ z = 2-t \end{array}\right\} t \in \mathbb{R}$

$3x-4y+z-17=0$.

$3(4+3t)-4(-1+2t)+(2-t)-17=0$

$12+9t+4-8t+2-t-17=0$

$0t=-1$

no solution.

$\therefore$ line does not lie in the plane.

3) $3x-2y-7z-6=0$

a) x-axis

$3x-2(0)-7(0)-6=0$

$3x-6=0$

$\boxed{x=2}$

$(2,0,0)$

b) y-axis

$-2y-6=0$

$\boxed{y=-3}$

$(0,-3,0)$

c) z-axis

$-7z-6=0$

$\boxed{z=\frac{-6}{7}}$

$(0,0,-\frac{6}{7})$

4a) $\vec{r} = (6,-4,3) + s(-2,4,7) + t(-7,6,-3)$, $s,t \in \mathbb{R}$

$\vec{n} = (-2,4,7) \times (-7,6,-3)$

$= (-54,-55,16)$

OR $= (54,55,-16)$

$\begin{matrix} -2 & 4 & 7 & -2 & 4 \\ -7 & 6 & -3 & -7 & 6 \end{matrix}$

$= (-12-42,-49-6,-12+28)$

$= (-54,-55,16)$

$\therefore 54x+55y-16z+D=0$

$54(6)+55(-4)-16(3)+D=0$

$D=-56$.

$\therefore 54x+55y-16z-56=0$.

x-axis

$54x-56=0$

$x=\frac{56}{54}$

$\boxed{x=\frac{28}{27}}$.

y-axis

$55y-56=0$

$\boxed{y=\frac{56}{55}}$

z-axis

$-16z-56=0$

$z=\frac{-56}{16}$

$\boxed{z=\frac{-7}{2}}$

5) $\vec{r} = (6,10,1) + t(3,4,-1)$, $t \in \mathbb{R}$

$$\left.\begin{array}{l} x = 6+3t \\ y = 10+4t \\ z = 1-t \end{array}\right\} t \in \mathbb{R}$$

a) xy-plane

$z = 0$

$1 - t = 0$

$t = 1$

P.O.I. $(9, 14, 0)$

b) xz-plane

$y = 0$

$10 + 4t = 0$

$4t = -10$

$t = \frac{-5}{2}$

P.O.I $\left(\frac{-3}{2}, 0, \frac{7}{2}\right)$

c) yz-plane

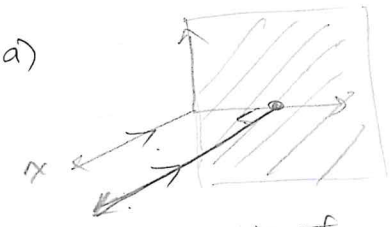
$x = 0$

$6 + 3t = 0$

$t = -2$

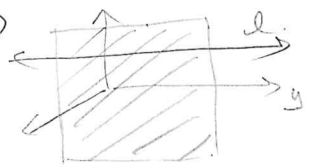
P.O.I. $(0, 2, 3)$

6 a)



one point of intersection.

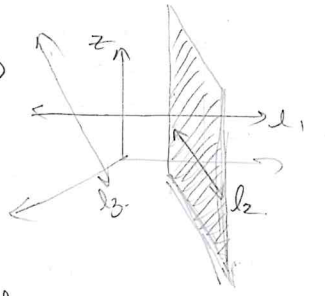
b)



i) either no intersection as parallel & distinct

OR ii) infinite solutions as line is on plane

c)



l_1 = one solution

l_2 = ∞ solutions

l_3 = $\emptyset$ solution

7) plane : $3x - 2y + 7z - 31 = 0$

line : $\vec{r} = (0,0,0) + t(3,-2,7)$, $t \in \mathbb{R}$

$$\left.\begin{array}{l} x = 3t \\ y = -2t \\ z = 7t \end{array}\right\} t \in \mathbb{R}$$

$3(3t) - 2(-2t) + 7(7t) - 31 = 0$

$9t + 4t + 49t - 31 = 0$

$62t = 31$

$t = \frac{1}{2}$

$\therefore$ P.O.I. $\left(\frac{3}{2}, -1, \frac{7}{2}\right)$

8) plane $4x-2y+5z+18=0$

line : $\vec{r} = (6,-2,-2) + t(4,-2,5)$ $t \in \mathbb{R}$

$$\left.\begin{aligned} x &= 6+4t \\ y &= -2-2t \\ z &= -2+5t \end{aligned}\right\} t \in \mathbb{R}$$

sub. $4(6+4t) - 2(-2-2t) + 5(-2+5t) + 18 = 0$

$$24+16t+4+4t-10+25t+18=0$$

$$45t = -36$$

$$t = \frac{-4}{5}$$

$\therefore$ POI $\left(\frac{14}{5}, \frac{-2}{5}, -6\right)$

11)

$$\frac{x-k}{3} = \frac{y+4}{2} = \frac{z+6}{1}$$

plane : $x-4y+5z+5=0$.

$$\left.\begin{aligned} x &= k+3t \\ y &= -4+2t \\ z &= -6+t \end{aligned}\right\} t \in \mathbb{R}$$

$$(k+3t) - 4(-4+2t) + 5(-6+t) + 5 = 0$$

$$k+3t+16-8t-30+5t+5=0$$

$$0t = k-9.$$

a) single point

→ since coefficient of t is zero, not possible

→ eg. $\vec{m} \cdot \vec{n}$

$(3,2,1) \cdot (1,-4,5)$

$= 3-8+5$

$= 0$ $\therefore$ Line // plane parallel

b) infinite # of points

if $0t = 0$.

$\therefore k-9=0$

$\boxed{k=9}$

c) no points

if $0t = H$, $H \neq 0$

$\therefore k-9 \neq 0$

$\boxed{k \neq 9}$